

Digital HPH HR 800 UPS (800VA)

Features

- True sinewave inverter
- Delivers utility-grade AC power to operate AC-powered electronic equipment like plasma TV sets and entertainment systems and appliances like lights, fans, refrigerators and washing machines
- Advanced microprocessor-based control
- Fast-action AC synchronised transfer switch for seamless changeover from main mode to inverter mode during power cuts
- LED display for overload, battery loose connection, inverter fault, short circuit and so on
- LCD display (optional)
- RS232 interface (optional)
- SNMP Web interface (optional)
- Audible alarm and many protection features



Price: ₹ 4200

do not exceed three years of lifetime as against five years in the case of tubular batteries. But the initial cost of tubular batteries is also higher.

SMF batteries that are available in the market today are generally not the correct solution for inverter application because inverter is a heavy-duty application and SMF batteries won't give a good life for such an application.

Some companies have designed SMF batteries with the characteristics of thick-, flat-plate batteries. These batteries give you two advantages: They don't require topping up and can last long enough for an inverter kind of application.

Tubular batteries come in normal-size, jumbo and tall containers. Capacities are generally 135 Ah, 150 Ah or 180 Ah. The jumbo and tall containers have extra electrolyte, so the topping-up frequency is reduced.

Which inverter-battery combination will work the best?

You may think that a combination of the best inverter brand and the

best battery brand will give you the best backup. But rather than buying inverter and battery separately, go for a package. Otherwise, you will be running after two companies if something goes wrong with the battery or inverter. The inverter maker will say it's a battery problem and the battery supplier will say it's an inverter problem. There is no one-point solution. So buy both the battery and the inverter of the same brand, or at least from the same vendor.

Given the high cost of the battery, the vendor should be able to take care of both—the inverter and the battery.

Efficiency

To convert power, some power is lost as heat and this is unavoidable. The efficiency of your inverter depends on the ratio of the amount of power generated to the amount of power that is lost. So it is better to buy an inverter that conforms to your power needs. If your power requirement is minimal and you buy a large inverter, the efficiency will be low, as the inverter

requires some power to run itself as well. Every inverter manufacturer stamps an efficiency graph (efficiency vs load) that you must consult carefully. Make sure that the curve starts from zero.

Also, the inverter consumes some power even when it is not running any load. So if you want to keep the inverter running for long hours, make sure its idle power is low. You can find the inverter's idle power described in the specifications sheet.

Safety

Don't go for cheap, unbranded models, as these are usually fraught with risk. Check whether the inverter has been certified by any testing organisation like ISI, UL and ETL. A safety certification stamp ensures that the inverter will be safe to use and shall also meet the manufacturer's specifications.

The sensitive portions of the inverter should be well protected against overload and excessive heating. Also, there must be protection from lighting surges. In case the inverter can't stand too high or too low a load, it must have the necessary mechanism to shut itself down completely.

Last but not the least

Some sellers are using the term 'digital inverter,' fooling the customers into believing that these are sinewave. The fact is these are purely analogue inverters and not really sinewave. To eliminate the confusion, ask for a DSP-based sinewave inverter or home UPS.

Also, as mentioned earlier, don't forget to consider the power factor conversion rating of the model being considered. For example, an 800VA inverter with a power factor of 0.7 would give an output of 560 watts, whereas another 800VA model with a power factor of 0.5 would give a maximum output of 400 watts. ●

The author is executive editor at EFM